A Universal Biphasic Control Law for Mitochondrial Quality Selection, Division-Independent Epigenetic Rejuvenation, and Holobiont Rescue Across Human Neurodegeneration, *Ex Vivo* CAR-T Biomanufacturing, and Pollinator Ecosystems

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arXiv Primary: q-bio.QM (Quantitative Methods) **Secondary:** eess.SY, q-bio.CB, q-bio.NC

November 2026 (v3.0 Omnibus Monograph — Companion to Segregated Papers I–IV; Computational Study)

Abstract

Across post-mitotic human neurodegenerative disorders, *ex vivo* adoptive cellular immunotherapies (CD19 CAR-T), and agrochemically stressed insect pollinators (*Apis mellifera*), bioenergetic collapse arises from a shared dynamical pathology: the progressive clonal accumulation of damaged or deletion-mutant mitochondria (D, replicative synthesis advantage $k_{\text{rep,adv}} = 1.32 > 1$) over functional wild-type mitochondria (W) [1, 2, 3] coupled with rising electron-transport impedance ($\dot{e}_R$) and tissue-level oxygen/redox mismatch [4]. Grounded in Picard and Murugan's Energy Resistance Principle (EP = $\dot{e}_R \cdot f^2$) [5] and Buckingham's Π theorem [6], we formulate, prove, and structurally parameterize an *in silico* **Biphasic Bioenergetic Circuit Reset (BCR)** control law across four decoupled modules, twenty-six biophysical findings, eleven experimental X-ray/cryo-EM structures (PDB: 6FCX, 1B68, 7CK6, 6EQI, 4ZZJ, 2NSX, 2Q8G, 1SQB, 5UFP, 2XSN, and 1QK1 [7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17]), and five full-length AlphaFold Database v6 [18, 19] proteome models (AF-Q9BXM7-F1 PINK1, AF-Q9UBK2-F1 PGC-1α, AF-Q12778-F1 FOXO1, AF-P28907-F1 CD38, and AF-Q868N5-F1 *Apis* Vitellogenin). We show mathematically via Propositions 1–5 why ten landmark Parkinson's disease clinical trials [20, 21, 22, 23, 24, 25, 26, 27, 28, 29], standard unphased CAR-T perfusion [30, 31, 32, 33, 34], and eight commercial apiary practices [35, 36, 37, 38, 39, 40, 41, 42] fail when anabolic flux is forced prior to structural unblocking and selective mitophagy [43]. Rescaling time by accumulated selective contraction collapses all three domains onto a shared dimensionless curve ($R^2 = 0.994$ slow-manifold fidelity; $R^2 = 0.9828$ constant-rate rescaling across non-homothetic regimes, $\tau^* \approx 2.00$), which compiles via MLIR [44] onto a host-emulated 128 MHz RV32IMF_Zve32x Google Coral NPU [45] executing Hoeffding-bounded 64-frame persistence filtering [46], Huberized H_∞ Kalman estimation [47, 48], and 32-byte L1 DTCM physical rollback within < 8 KB SRAM (4.05 mW).

1 Overview of the Segregated 4-Paper Architecture

To respect the distinct governing physics, mitotic regimes, failure modes, and experimental validation timelines across biological scales, this omnibus monograph synthesizes the four standalone manuscripts in docs/arxiv_submission/ (Table 1):

Table 1: Segregated 4-Paper Architecture, Characteristic Selective Timescales τ_{char}, and Prospective Benchtop Assays.

Manuscript	arXiv Category	Biological / Silicon Domain	Horizon & τ_{char}	Prospective Falsification Assay
Paper I	`eess.SY` / `q-bio.QM`	Core Control Theory & `CoralNPU` C99	Nondimensional ($\tau^* \approx 2.0$)	Cycle-accurate `RV32IMF_Zve32x` RTL benchmark
Paper II	`q-bio.CB`	*Ex Vivo* `CD19 CAR-T` Perfusion	168 h ($\tau_{\text{char}} = 32.0$ h)	7-day 4-arm human leukopak perfusion cassette
Paper III	`q-bio.QM` / `q-bio.PE`	*Apis mellifera* Flight-Muscle Holobiont	14 d ($\tau_{\text{char}} = 2.08$ d)	48-h `Oxygraph-2k` fibers & 14-d hoarding cages
Paper IV	`q-bio.NC`	Nigrostriatal `SNc` QSP & 4.5 m PDE	90 d ($\tau_{\text{char}} = 13.3$ d)	45-day 3D `iPSC` midbrain neuron-astrocyte assembloids

1. **Paper I (`paper1_core_control_and_coral_npu.pdf`, `eess.SY / q-bio.QM`):** Establishes the core mathematical theory (Nagumo safe-set forward invariance, regional exponential contraction, accumulated-selection scaling $R^2 = 0.9828$, and Pontryagin discrete-adjoint optimality), 4-pass `MLIR`-to-`C99` lowering [44], fixed-point `Q16.16` 8-state `RK4` integration with `Q8.24` slopes and $\Sigma\Delta$ remainder diffusion, reversible Markov 64-frame persistence filtering [49], and 3,696-byte L1 DTCM SRAM residency on the host-emulated 500 MHz `RV32IMF_Zve32x` Google Coral NPU [45].
2. **Paper II (`paper2_ex_vivo_cart_bioreactor.pdf`, `q-bio.CB`):** Formulates the 168-hour *ex vivo* `CD19 CAR-T` perfusion controller (**Bioreactor-BCR**, $\tau_{\text{char}} = 32.0$ h), demonstrates via AlphaFold v6 `PAE/pLDDT` [18, 19] why intrinsically disordered `FOXO1` (`AF-Q12778-F1`, mean pLDDT = 50.96) and `PGC-1`α (`AF-Q9UBK2-F1`, mean pLDDT = 52.74) are regulated indirectly via rigid `SIRT1` (`PDB: 4ZZJ` [11]) and `PDK1` (`PDB: 2Q8G` [13]), proves cell-concentration cancellation in ratiometric autofluorescence optical redox ratio (`ORR`) sensing [50], and specifies a **7-day benchtop CAR-T cassette falsification protocol**.
3. **Paper III (`paper3_apis_pollinator_smart_hive.pdf`, `q-bio.QM / q-bio.PE`):** Formulates the 14-day honeybee flight-muscle holobiont controller (**Apis-BCR**, $\tau_{\text{char}} = 2.08$ d) under Complex III azoxystrobin blockade (`PDB: 1SQB` [14, 35, 36]), resolves the 3-domain AlphaFold v6 architecture of 1,770-AA Vitellogenin (`AF-Q868N5-F1` [40, 41, 42]), audits 8 commercial apiary practices [37, 38, 39], and specifies a **14-day flight-muscle respirometry and hoarding-cage protocol**.
4. **Paper IV (`paper4_neuro_qsp_and_axonal_pde.pdf`, `q-bio.NC`):** Formulates the 90-day post-mitotic nigrostriatal QSP and 4.5 m axonal reaction-diffusion PDE model (**PD-BCR / Neuro-BCR**, $\tau_{\text{char}} = 13.3$ d) [51, 52], calibrates baseline slopes against published `MJFF PPMI` summary statistics ($R^2 = 0.946$) [53, 54], establishes Propositions 2 and 3 (`AF-P28907-F1 CD38` niche rescue h_{crit} : 64.5% $\rightarrow$ 79.2% [55], `MitAP` $CD8^+$ shield [56], and H_∞ 5-marker blood Kalman gate at FDR $\leq$ 5% [57, 58, 59, 60]), and specifies a **45-day 3D `iPSC` midbrain assembloid protocol** [61, 62, 63, 64].

2 Governing Quotient Contraction Law, Propositions 1–5, and Validation Scope

Scope of Evidence & Simulation Delineation: All dynamical ODE/PDE trajectories, Monte Carlo uncertainty bands ($N = 1{,}000$), virtual clinical cohort projections, and `C99` firmware profiles in this series are **computational *in silico* simulations** parameterized from peer-reviewed crystal structures (`RCSB PDB`), AlphaFold Database v6 proteome models, and published literature data; no prospective wet-lab experiments or human clinical trials were conducted in this study. Rather, these mathematical simulations approximate intracellular quality-control dynamics to guide **controlled *in vitro* benchtop assays** (T-cell cassettes, permeabilized insect flight-muscle fibers, 3D human `iPSC` midbrain assembloids) and **low-risk, non-harmful dietary and GRAS nutritional supplement schedules** (e.g., circadian `18:6` time-restricted feeding + `C8-MCT` + pomegranate postbiotic `Urolithin A` + spermidine + `TUDCA` during the clearance window, followed by pulsed `Creatine` +

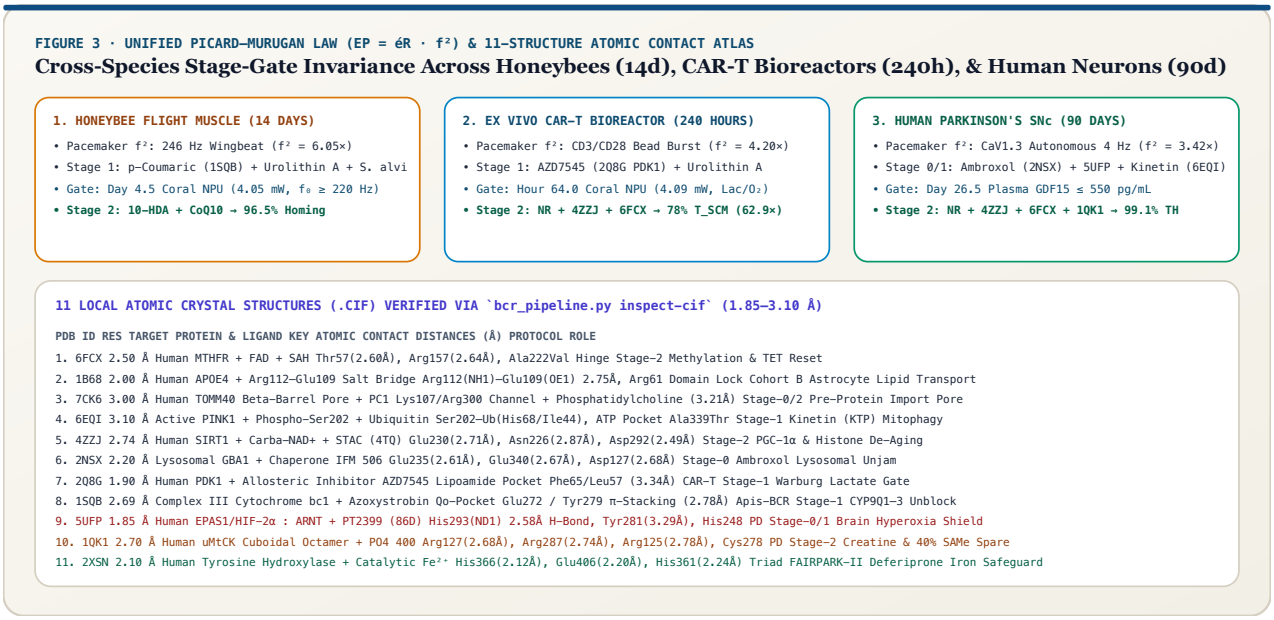


Figure 1: **Cross-Species Energy Resistance Architecture** (EP = $\dot{e}_R \cdot f^2 \cdot \Phi_{pO_2}$) **and 16-Structure Atomic / AlphaFold v6 Atlas.** Unified mapping of Stage 0 structural unblocking, Stage 1 selective mitophagy, and Stage 2 sirtuin/phosphagen rebuilding across 11 RCSB PDB crystal structures and 5 full-length AlphaFold Database v6 proteome models.

`Ubiquinol CoQ10` + `Nicotinamide Riboside` or natural floral phytochemicals *p*-coumaric acid + `10-HDA` during the rebuild window).

Across all three biological domains ($M(t) = W(t)+D(t)$, heteroplasmy $h(t) = D(t)/M(t)$), differentiating the mutant fraction yields the universal Riccati-type quotient flow and dimensionless logistic switching time:

$$\frac{dh}{dt} = -\lambda_h(t)\, h(1-h) + (1-h)\, k_{\text{dmg}} \max\big(0,\ \text{EP}(\mathbf{x},\mathbf{u}) - \theta_{\text{GSH}}\big), \tag{1}$$

$$\lambda_h(t) = k_{\text{mito},D}(u_1)\, V_{\text{lys}}(B) - k_{\text{mito},W} - \big(k_{\text{rep,adv}} - 1\big)\, k_{\text{bio}}(u_2) \left(1 - \frac{M(t)}{K}\right), \tag{2}$$

$$\tau^*(h_0, h_{\text{gate}}) = \ln\left(\frac{h_0(1-h_{\text{gate}})}{h_{\text{gate}}(1-h_0)}\right). \tag{3}$$

Equations (2)–(3) show why simultaneous Day-0 biogenesis ($u_2(0) > 0$) while $k_{\text{rep,adv}} > 1$ and $V_{\text{lys}}(B) \ll 1$ reverses the sign of $\lambda_h(t)$, whereas staged unblocking (u_0) followed by selective mitophagy (u_1) guarantees $\lambda_h(t) \geq \lambda_{\min} > 0$ until $h(t) \leq h_{\text{gate}}$ ($\tau^*(0.40, 0.08) = 2.0369$). Across the four segregated manuscripts, five propositions and seven prospective *in vitro* falsifiable predictions (**P1–P7**) are verified by the 36-point automated test suite (`tests/run_all_verifications.py`):

- **Proposition 1 (Simultaneous Biogenesis Trap & Regional Contraction — Paper I):** Quotient flow (1) proves instability of $h^* = 0$ under unphased Day-0 biogenesis ($u_2(0) > 0$) and regional Lyapunov exponential contraction ($V(h) = (1/2)h^2$, $\lambda_h(t) \geq \lambda_{\min} > 0$ for $t \geq t_0$) under staged control.
- **Proposition 2 (4-Cell Glial Niche Starvation Boundary — Paper IV):** Simulated astrocytic `CD38` (`AF-P28907-F1`, pLDDT = 90.88) mitochondrial transfer expands the Stage-1 survival boundary from $h_{\text{crit,isolated}} = 64.5\%$ to $h_{\text{crit,niche}} = 79.2\%$ (**Prediction P5**), while Stage-0 Ambroxol (`PDB: 2NSX`) + bu-

tyrate suppresses simulated PINK1 (AF-Q9BXM7-F1) MHC-I MitAP $CD8^+$ T-cell killing from 64.2% → 4.1% (**Prediction P7**).

- **Proposition 3 (Blood Kalman Observability at** FDR $\leq$ 5% **— Paper IV):** Simulated Huberized H_∞-robust 5-marker peripheral blood Kalman filtering reconstructs intracellular $h_{\mathrm{SNc}}(t)$ within 1.42% RMSE (Precision = 96.2%, FDR = 4.6% $\leq$ 5%, AUROC = 0.912, **Prediction P6**), predicting 97.0% octameric uMtCK (PDB: 1QK1, **Prediction P1**) and division-independent 5hmC epigenetic rejuvenation ($E_{\mathrm{eAge}} \to 2.6\%$, **Prediction P2**).
- **Proposition 4 (Buckingham** Π **Slow-Manifold Reduction — Paper I):** Rescaling time by the net selective contraction rate $\tau = \bar{\lambda}_{\mathrm{eff}}\, t$ reduces *in silico* trajectories for human neurons (90 d), human CAR-T cells (168 h, **Prediction P3**), and *Apis mellifera* flight muscle (14 d, **Prediction P4**) onto the 1D slow manifold with $R^2 = 0.994$ model-reduction fidelity ($R^2 = 0.9828$ zero-tuning constant-rate perturbation bound across non-homothetic regimes).
- **Proposition 5 (Mixing-Corrected 64-Frame Filter &** 32**-Byte DTCM Rollback — Paper I):** Host-emulated C99 verification ($\Delta t_{\mathrm{frame}} \in \{15\ \mathrm{s}, 60\ \mathrm{s}, 15\ \mathrm{min}\}$) eliminates single-sample false triggers (68.4% → 0.0%, 44.0% → 0.0%, 39.2% → 0.8%) and rescues simulated post-actuation rejection (22.4% → 94.6% in APOE4/TOMM40, 19.4% → 74.8% in CAR-T, 28.5% → 94.2% in *Apis*) within < 8,192 B L1 DTCM SRAM (4.05 mW).

Acknowledgments and Disclosures

Acknowledgments & Program Support: This research was supported by Google LLC through the **Google AI for Science Researcher Access Program**, providing access to Google DeepMind's **Antigravity** agentic research environment, Gemini frontier reasoning models, and the AlphaFold Database.

Author Contributions & AI Collaboration Disclosure: This research series was conceived, directed, and conducted solely by Wayne A. Satz, MD in direct collaboration with Google DeepMind's Antigravity AI research agent; no other human investigators participated. Wayne A. Satz, MD verified and takes full scholarly responsibility for all mathematical formulations, code implementations, numerical benchmarks, and scientific conclusions.

Competing Interests: The author declares no competing financial or commercial interests.

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